

Math 54 Discussion Worksheet

Name: _____ ID: _____

Problem 1

Let H and K be subspaces of a vector space V . The intersection of H and K , written as $H \cap K$ is the set of $v \in V$ such that $v \in H$ and $v \in K$. Show that $H \cap K$ is a subspace of V . Then, give an example in $\mathbb{R}^2$ to show that the union of two subspaces is not generally a subspace.

Problem 2

Given subspace H and K of a vector space V , define $H + K$ to be the set of all vectors $v \in V$ that can be expressed as the sum of a vector in $h \in H$ and a vector $k \in K$, i.e.

$$H + K = \{v \in V : v = h + k \text{ for } h \in H, k \in K\}.$$

- a) Show that H is a subspace of $H + K$ and that K is a subspace of $H + K$.
- b) Show that $H + K$ is a subspace of V .

Problem 3

Recall that the nullspace of a matrix A is the set of solutions to $Ax = 0$. Also, the column space of A is the span of the columns of A . Show that for an $m \times n$ matrix A , $\text{Null}(A)$ is a subspace of $\mathbb{R}^n$ while $\text{Col}(A)$ is a subspace of $\mathbb{R}^m$. Describe in words (or by clear, generalizable examples) procedures to determine the nullspace and column space of a matrix A .